SID for DNS

`draft-sid-dns-encoding.md`

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Objective

- Why:
  - Need to locate a .sid file to discover coreconf structure
  - DNS: exists and is robust
- How:
  - Define a **technical** way to store SID into the DNS
  - Propose an administrative delegation for the resolution
  - Need a global postfix `.sid.yt` but `.sid.iana.` targeted
- Principle:
  - full 63 representation, then reverse (like reverse IPv6):
  - entry_point contains module information (i.e. ref to .sid file)
  - other sids points to entry_point.

```
canonical (20 digits):
  0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 5 5 0
  <-- most significant             least -->

reversed (SID name):
  0 5 5 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

fully qualified SID DNS name:
  0.5.5.2.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.sid.yt.
```

SID

- 63 bits positive allocated to YANG identifiers
- Pool managed by IANA:
  - 0 - 999 999 : IETF
  - 1 - ….. : other SDOs (and may be registries) ?
  - 3 000 000 000 : PEN
  - 300 000 000 000 : PEN
- Constrains:
  - Once assigned to an identifier, cannot be changed (just obsoleted)
  - point to a
-

Resource Records

- Add TXT
- `entry_point` SID contains main information
  - i.e. url of the .sid file
  - /!\ NEED A STABLE REPOSITORY
- other SIDs of the module leads to the `entry_point`

```
$TTL 300
@   IN  SOA  ns1.sid.yt. hostmaster.sid.yt. (
        2026060301  ; Serial
        3600        ; Refresh
        1800        ; Retry
        604800      ; Expire
        300 )     ; Minimum TTL

@   IN  NS   ns1.sid.yt.
@   IN  NS   ns6.gandi.net.

; --- coreconf-m2m (entry-point 100000) ---

; SID 100000  [module] coreconf-m2m
0.0.0  IN  TXT  "name=coreconf-m2m"
0.0.0  IN  TXT  "repository=http://user.plido.net/SIDs/coreconf-m2m@2026-05-26.sid"

; SID 100001  [identity] air-temperature
1.0.0  IN  TXT  "name=coreconf-m2m"
1.0.0  IN  TXT  "entry_point=100000"

; SID 100002  [identity] average-distance
2.0.0  IN  TXT  "name=coreconf-m2m"
2.0.0  IN  TXT  "entry_point=100000"

; SID 100003  [identity] barometric-pressure
3.0.0  IN  TXT  "name=coreconf-m2m"
3.0.0  IN  TXT  "entry_point=100000"

; SID 100004  [identity] east-wind-speed
4.0.0  IN  TXT  "name=coreconf-m2m"
4.0.0  IN  TXT  "entry_point=100000"

; SID 100005  [identity] north-wind-speed
5.0.0  IN  TXT  "name=coreconf-m2m"
5.0.0  IN  TXT  "entry_point=100000"

; SID 100006  [identity] precipitation
6.0.0  IN  TXT  "name=coreconf-m2m"
6.0.0  IN  TXT  "entry_point=100000"
```

Usage

```
--- CBOR (hex) ---
a11a000186df92a418231a000186a80100130118246457 2f6d32a418231a000186a60100130318246 26d6da418231a000186a90100130018246 0a418231a000186a20100130118246 26b6da418231a000186ad0100130
636b5061a418231a000186a30100130218246 36b5061a418231a000186a70100130118246 3255248a418231a000187040100130118246 4646656743a418231a000187050100130118246 3646567a418231a000186b0010

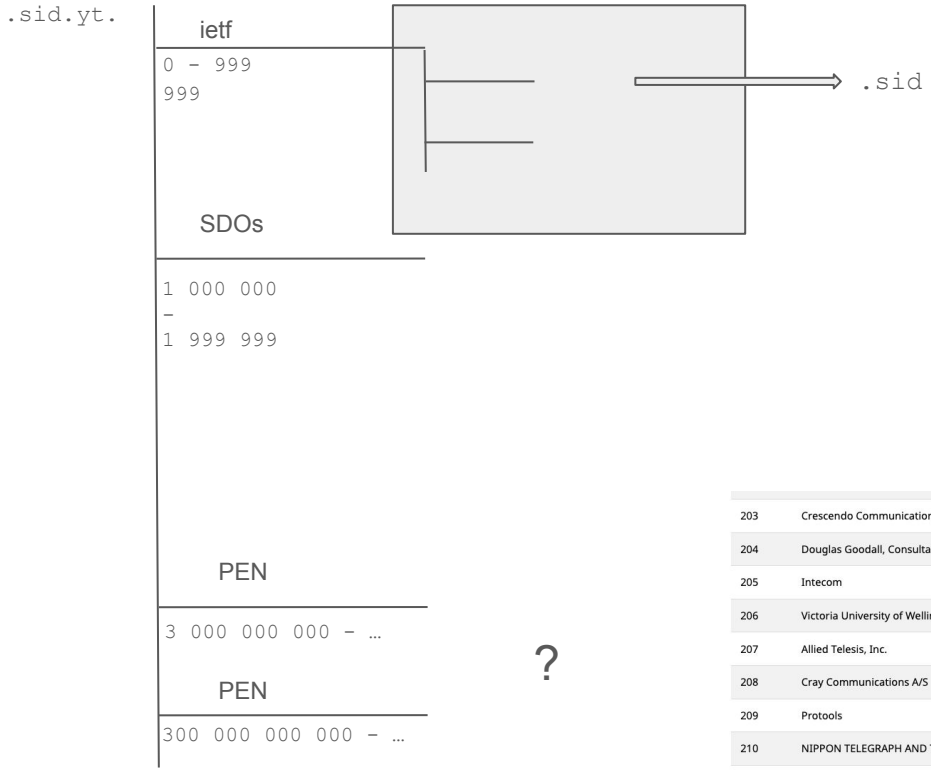
--- JSON ---
[DEBUG DNS] SID=100063 module=coreconf-m2m entry_point=100000 repository=http://user.plido.net/SIDs/coreconf-m2m@2026-06-07.sid
{"coreconf-m2m:transducers/transducer": [{"type": "coreconf-m2m:solar-radiation", "id": 0, "precision": 1, "unit": "W/m2"}, {"type": "coreconf-m2m:precipitation", "id": 0, "
 {"type": "coreconf-m2m:wind-direction", "id": 0, "precision": 1, "unit": "deg"}, {"type": "coreconf-m2m:wind-speed", "id": 0, "precision": 2, "unit": "m/s"}, {"type": "core
cision": 1, "unit": "degC"}, {"type": "coreconf-m2m:vapor-pressure", "id": 0, "precision": 3, "unit": "kPa"}, {"type": "coreconf-m2m:barometric-pressure", "id": 0, "precisio
nit": "degC"}, {"type": "coreconf-m2m:compass-heading", "id": 0, "precision": 1, "unit": "deg"}, {"type": "coreconf-m2m:x-orientation", "id": 0, "precision": 1, "unit": "deg
2m:east-wind-speed", "id": 0, "precision": 2, "unit": "m/s"}]}
```

Use pycoreconf:

- pycoreconf convert JSON <->CBOR based on extended .sid file
- sid file extension is under the adoption process at IETF
- proposition: use DNS to automatically find the .sid file

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	54.37.158.10	213.186.33.99	DNS	122	Standard query 0x0e76 TXT 2.6.0.0.0.1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.ietf.s…
2	0.136691671	213.186.33.99	54.37.158.10	DNS	183	Standard query response 0x0e76 TXT 2.6.0.0.0.1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0…
3	0.163822377	54.37.158.10	213.186.33.99	DNS	122	Standard query 0x62bb TXT 0.0.0.0.0.1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.ietf.s…
4	0.306408400	213.186.33.99	54.37.158.10	DNS	230	Standard query response 0x62bb TXT 0.0.0.0.0.1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0…
5	0.342601746	54.37.158.10	213.186.33.99	DNS	85	Standard query 0x7bce A user.plido.net OPT
6	0.342721363	54.37.158.10	213.186.33.99	DNS	85	Standard query 0xd296 AAAA user.plido.net OPT
7	0.367470815	213.186.33.99	54.37.158.10	DNS	148	Standard query response 0xd296 AAAA user.plido.net CNAME vps-722eeb27.vp…

Proposal



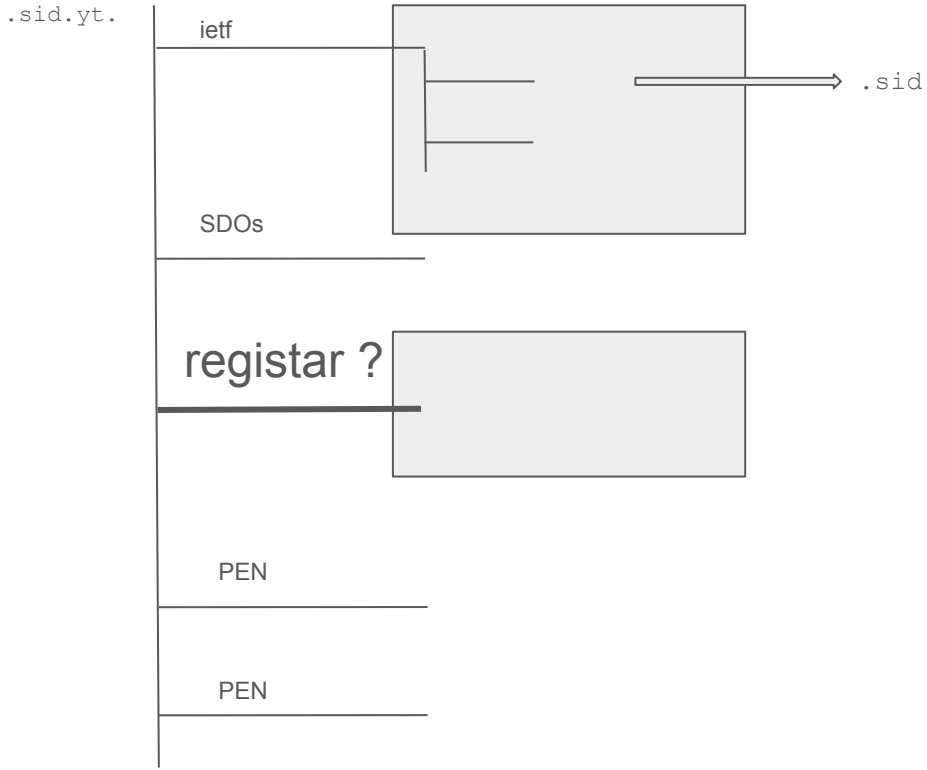
?

A CORECONF serialization may be composed of several YANG DM. Some comes for standards, other from manufacturers.

203	Crescendo Communication, Inc.	Prem Jain	prem@cres.com
204	Douglas Goodall, Consultant	Douglas Goodall	doug@goodall.com
205	Intecom	Patrick Deloulay	pdelou@intecom.com
206	Victoria University of Wellington	Laurie Ellims	laurie.ellims@vuw.ac.nz
207	Allied Telesis, Inc.	Hermin Anggawijaya	hermin.anggawijaya@alliedtelesis.co.nz
208	Cray Communications A/S	Hartvig Ekner	hj@craycom.dk
209	Protools	Glen Arp	---none---
210	NIPPON TELEGRAPH AND TELEPHONE CORPORATION	Tatsuya Miyagi	netadmin@ml.hco.ntt.co.jp
211	Fujitsu Limited	Kei Nakata	nakata.kei@fujitsu.com

- 0 - 999 999

Proposal



registrars may allocate SID ranges, maintain in time .sid files and integrity.

Open questions

- Is there a need for non-standard allocated SID ?
- How to guarantee consistency in resolution ?
- Is DNSSEC mandatory to secure the resolution ?